

Helping or Hindering? User Experience Trade-offs in Refinement-Based AI Moderation for Online Comments

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Abstract

Toxic and aggressive comments are a persistent problem in online spaces, and platforms commonly rely on blocking-based moderation to address them. However, blocking is only one way of intervening during comment writing, and alternative approaches that aim to preserve user expression have rarely been examined in direct comparison. This study aims to compare blocking and refinement-based moderation during comment writing to assess how they differentially shape user behavior, user perceptions, and semantic properties of the resulting comments. We deployed a web-based commenting platform and conducted a controlled user experiment in which moderation intervened in real time. Our results show that refinement primarily affects the semantic and emotional characteristics of comments, whereas blocking influences users' evaluative responses to the moderation experience. These findings clarify that moderation strategies intervene in different dimensions of the commenting process and provide a basis for designing moderation systems that balance textual outcomes and user experience.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Collaborative and social computing systems and tools*; • **Computing methodologies** → *Natural language processing*.

Keywords

Online content moderation; toxicity detection; polite rewriting; refinement-based moderation; NLP; human-centered AI

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1 Introduction

Online comment spaces play an important role in everyday communication on digital platforms, but they also frequently become sites of hostile or aggressive expression. In response, platforms have increasingly relied on automated moderation systems that

detect and block toxic comments [3]. While effective at scale, such systems typically operate through a binary enforcement logic: once a comment is classified as problematic, it is removed. This framing positions moderation as a decision about publication outcomes, leaving open questions about how alternative forms of intervention shape the experience of expression during writing.

While techniques for rewriting or transforming text have been explored in machine learning research, they have rarely been examined in the context of comment moderation. In particular, little is known about how rewriting-based interventions are perceived when introduced as part of the comment writing process itself.

In this study, we take a comparative approach to blocking-based and refinement-based moderation within a real-time comment writing interface. We examine users' evaluations of the moderation experience and the expressive characteristics of their comments to explore how different intervention strategies operate during writing.

2 Related Work

2.1 Post-hoc Enforcement in Comment Moderation

In online comment moderation, blocking is commonly implemented as a form of post-hoc enforcement applied to already written comments. At the system level, automated moderation mechanisms typically assess completed text and intervene by flagging, filtering, or deleting content based on detected violations [3]. Research on enforcement-based comment moderation has largely focused on how users interpret and evaluate moderation decisions after enforcement, including judgments of fairness, trust, transparency, and acceptance [1, 5, 10]. These studies suggest that enforcement-based moderation can increase perceived credibility and acceptance when users evaluate moderation outcomes as legitimate and transparent. Because these evaluations occur post hoc, they provide limited insight into how moderation is encountered during the writing process. By contrast, this study examines moderation as an intervention encountered during the act of writing.

2.2 Polite Rewriting and Text Style Transfer

Polite rewriting has primarily been studied within the literature on text style transfer, where the goal is to modify the manner of expression while preserving underlying semantic content [6]. Subsequent work has shown that sequence-to-sequence architectures trained on parallel impolite-polite data can generate fluent rewritten text, including models based on BART and related encoder-decoder frameworks [8, 11]. While these works demonstrate the technical feasibility of polite rewriting, they offer limited insight into how



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rewriting functions as an interactive intervention within real-time comment writing interfaces. This study builds on prior work by examining how polite rewriting functions as an interactive intervention within real-time comment writing, rather than as a standalone text transformation.

2.3 Moderation Strategies in Comparison

Blocking-based moderation operates through binary enforcement decisions that permit or prevent publication, leaving users with limited opportunity to revise their expression at the moment of intervention. Although moderation decisions in practice rely on contextual judgment and ambiguity [12], these complexities are rarely reflected in user-facing moderation interfaces. As a result, users often encounter moderation only through its final outcome or infer its presence from indirect cues rather than through explicit feedback [4]. In this framing, moderation is experienced primarily as a decision about publication. By contrast, alternative approaches such as refinement-based rewriting intervene during writing and aim to support revision, adopting a different interactional logic.

Despite these conceptual differences, prior research has rarely examined blocking-based enforcement and refinement-based rewriting within the same interactional context of comment writing. It therefore remains unclear how these approaches differ when they intervene at the same moment during comment writing, and how users experience these interventions as part of the writing process. This study addresses this gap by comparing blocking- and refinement-based moderation within the same real-time writing interface, to examine how these interventions are experienced by users during comment writing.

3 Methods

3.1 System Overview

We compare how two moderation strategies shape users' writing experiences and the resulting characteristics of comments when intervention occurs in real time. To enable this comparison, we developed a web-based commenting platform to examine how people experience AI-based moderation during the act of writing. For each comment, the system computes a continuous toxicity score (logit) and intervenes only when the score exceeds a predefined threshold.

3.2 Models

To support real-time interaction, toxicity estimation is separated from language refinement, allowing the system to determine when intervention is needed and how expression can be supported. For toxicity estimation, we employed a fine-tuned KoELECTRA model, following prior work that has widely adopted BERT-family classifiers for toxic language detection in online moderation settings, including Korean-language contexts [3]. For language refinement, we used a KoBART-based sequence-to-sequence model, consistent with prior work [8, 11]. Both models were trained on parallel toxic-polite sentence pairs, including augmented data constructed using GPT-based prompting, following established approaches in prior polite rewriting research [8, 9].

3.3 Experimental Design

We employed a between-subjects design with seven experimental conditions: blocking-based moderation and refinement-based moderation, each tested at three toxicity thresholds ($\tau = 0.3, 0.6, 0.9$), and a baseline condition with no intervention. A total of $N = 50$ participants were randomly assigned to one of the seven conditions (blocking: $n = 23$; refinement: $n = 19$; baseline: $n = 8$). Thresholds were selected to represent low, medium, and high levels of intervention intensity to examine whether the intensity and frequency of real-time intervention shape users' reactions and perceived intrusiveness.

In the blocking condition, comments exceeding the threshold could not be submitted and participants were prompted to revise their text without guidance. In the refinement condition, the system suggested a polite rewritten version; participants could accept or edit it. If the edited comment still exceeded the threshold, the system automatically submitted the suggested rewrite to prevent infinite revision loops. Participants completed three discussion tasks involving socially debated topics commonly encountered in Korean online comment spaces, producing nine comments in total.

To assess users' subjective responses to AI-based moderation, we administered a post-task survey grounded in the Technology Acceptance Model (TAM), following prior work applying TAM to AI-driven systems [2]. The survey assessed participants' evaluations of the moderation system, including usability, overall attitudes toward the moderation strategy, and perceptions of communication quality, freedom of expression, trust, and fairness.

3.4 Outcome Measures

3.4.1 Post-task Survey. The post-task survey was used to capture participants' subjective experiences with the moderation system. Perceived ease of use (PEOU) measured how easily participants could understand and interact with the system during comment writing, including the perceived complexity and burden of the writing process. Perceived usefulness (PU) assessed whether the moderation strategy was experienced as helpful for expressing opinions and composing comments. In addition to these TAM questions, the survey included items measuring the overall attitudes of participants toward the moderation policy, perceived communication quality, perceived freedom of expression, trust in the moderation system and perceived fairness of the intervention. Internal consistency was assessed using Cronbach's alpha, calculated separately for each condition. These measures capture users' evaluations of both the writing experience and the moderation policy itself.

3.4.2 Text-based Measures. We analyzed the submitted comments to examine how moderation strategies shaped communicative outcomes during writing. Sentiment analysis was conducted using a KoELECTRA-based model, which outputs a binary sentiment label (positive: 0 vs. negative: 1) and a continuous score indicating how clearly sentiment is expressed. The original comment text was used directly as model input without additional linguistic preprocessing.

4 Results

4.1 Behavioral Outcomes

Survey measures demonstrated acceptable internal consistency across conditions (see Appendix), supporting comparisons across moderation policies. We observed systematic differences in user perceptions across moderation policies, which we assessed using Welch’s *t*-tests for each survey measure. For overall satisfaction toward the moderation policy, the mean score under the blocking condition ($M = 3.875$, $SD = 0.248$, $n = 8$) was significantly higher than that under the refinement condition ($M = 3.378$, $SD = 0.785$, $n = 15$), $t = -2.250$, $p = .0368$. This result indicates that participants evaluated the blocking strategy more positively overall than the refinement strategy.

A similar pattern was observed for perceived communication quality. The blocking condition ($M = 4.042$, $SD = 0.603$) received significantly higher ratings than the refinement condition ($M = 3.417$, $SD = 0.724$), with the difference reaching statistical significance ($t = -2.205$, $p = .0416$).

In contrast, no statistically significant differences were found for PEOU ($p = .7127$), PU ($p = .1463$), perceived freedom of expression ($p = .7357$), perceived trust and fairness ($p = .5624$), and preference for group comparison ($p = .8288$).

4.2 Sentiment Outcomes of User Comments

4.2.1 Sentiment Label Analysis. Using the binary sentiment labels produced by the KoELECTRA model, we analyzed how the distribution of positive and negative comments varied across moderation conditions. Across the dataset, positive sentiment was higher than negative sentiment, with an approximate difference of 6.6 percentage points.

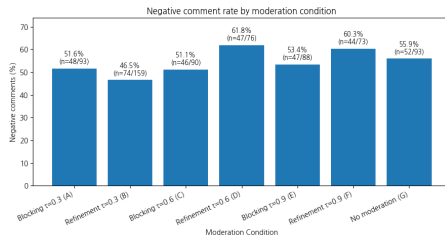


Figure 1: Negative comment rate by moderation condition ($n = \text{negative} / \text{total comments}$).

Figure 1 compares sentiment label distributions by group. Across the $\tau = 0.6$ and $\tau = 0.9$ conditions, refinement-based moderation exhibited a lower proportion of negative comments than blocking-based moderation and the baseline condition. Under the lower threshold ($\tau = 0.3$), however, interventions were triggered more frequently, and the sentiment distribution did not follow the same pattern. An analysis of variance (ANOVA) revealed significant differences in sentiment distributions between blocking-based and refinement-based conditions for both negative sentiment ($p = .027$) and positive sentiment ($p = .033$).

4.2.2 Sentiment Score Analysis. Beyond discrete sentiment labels, we analyzed sentiment scores, which indicate how strongly positive or negative an expression is, to examine differences in how sentiment was expressed across moderation conditions.

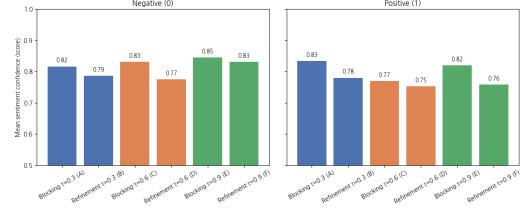


Figure 2: Mean sentiment score by group for each sentiment label

Figure 2 illustrates mean sentiment scores by group for each sentiment label. Across both positive and negative labels, refinement-based moderation consistently exhibited higher sentiment scores than blocking-based moderation. Across all thresholds, refinement-based moderation showed higher mean sentiment scores than blocking-based moderation for both sentiment labels.

5 Discussion

Our results show that refinement-based moderation improved the semantic properties of comments, whereas blocking-based moderation received higher ratings in overall satisfaction and perceived communication quality.

5.1 Refinement: Outcome Gains, Experience Costs

Across conditions, refinement-based moderation produced comments with fewer negative expressions and reduced emotional intensity compared to blocking. At the same time, participants evaluated the refinement experience less positively. In the experimental interface, refinement intervened repeatedly during the writing process and required participants to review, evaluate, and select suggested rewrites. Participants reported that repeated interactions increased perceived effort and disrupted the writing process. Consistent with prior findings that moderation experiences are shaped by how interventions are encountered rather than by outcomes alone [5, 10], refinement improved discourse outcomes but made the writing experience more demanding.

5.2 Blocking: Restriction without Friction

Blocking-based moderation produced a different pattern in the experiment. Although blocking constrained expression by preventing submission of certain comments, participants reported higher overall satisfaction and perceived communication quality. Unlike refinement, blocking occurred at a single, clearly defined moment and did not require participants to engage in repeated decision-making. This pattern can be understood through predictability and rule visibility in moderated spaces [7]. In our experiment, blocking concentrated the cost at a single point, allowing participants to revise or disengage without prolonged interaction. As a result,

blocking did not improve discourse quality, but resulted in a more stable and predictable moderation experience.

5.3 Two Dimensions of Moderation

Refinement and blocking intervened along different dimensions of moderation. The observed trade-off therefore reflects tension across dimensions—between improving discourse quality and maintaining a manageable moderation experience—rather than choices along a single continuum of moderation strength. This perspective suggests that effective moderation design should consider not only which outcomes to improve, but also when and how experiential costs are imposed on users during writing.

5.4 Limitations and Implications

This study focused on short-term interactions with a single moderation policy and relied on post-hoc self-report measures, limiting insight into longer-term learning, adaptation, and fine-grained writing processes. Moreover, the refinement-based condition introduced additional interaction steps compared to blocking-based moderation, which may have influenced user experience outcomes. In addition, some survey-based measures showed lower internal consistency in the blocking condition, due to the small sample size and response variability. Future work could examine longer-term use and richer process-level data with larger samples to better capture learning and adaptation over time. Despite these limitations, this study contributes by comparing blocking-based and refinement-based moderation within the same real-time writing interface. The findings highlight how the timing and interactional form of intervention shape users' experiences. Building on this distinction, this work opens opportunities for further discussion on moderation as a multi-dimensional design space, where strategies differ not only in outcomes but also in their interactional form during writing.

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Appendix

Table 1: Internal consistency of survey scales by condition (Cronbach's alpha)

Scale	Condition	Alpha	<i>k</i>	<i>n</i>
Attitude toward the Moderation Policy	Refine	0.859	3	15
	Block	0.581	3	8
Perceived Ease of Use (PEOU)	Refine	0.720	5	15
	Block	0.573	5	8
Perceived Usefulness (PU)	Refine	0.867	7	15
	Block	0.054	7	8
Behavioral Intention to Use	Refine	0.856	5	15
	Block	0.800	5	8
Perceived Freedom of Expression	Refine	0.733	5	15
	Block	0.845	5	8
Trust in the Moderation System	Refine	0.874	4	15
	Block	0.228	4	8
Perceived Communication Quality	Refine	0.747	4	15
	Block	0.721	3	8
Comparative Preference between Strategies	Refine	0.268	6	15
	Block	0.777	6	8